# What Has TabPFN's Prior Learned?

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## Progress Since the Proposal

- **Foundation (Kiet).** Built the shared repository: a data format for all runs, corrected metrics, nested synthetic datasets, a resumable sweep runner for local and Colab use, and a test suite.
- **Metric test (Kiet).** On surfaces of known shape, the proposal's step score S gives a smooth ramp and a sharp edge the same score (1.00 at 0°, 0.71 at 45°). Our new jump share separates them (0.00 vs. 0.98 at 0°; 0.00 vs. 0.96 at 45°).
- **Classic sweep (Nhan).** All 8,960 synthetic runs are complete: 8 datasets, 4 variants, 4 context sizes, 10 seeds, 7 models. Mean jump share: logistic regression 0.00, GP 0.03, SVM 0.04, random forest 0.71, MLP 0.73, gradient boosting 0.98, k-NN 1.00. Random forest and MLP are close on jump share but differ on axis alignment (0.94 vs. 0.79), so we read the two metrics together. An earlier 84-run rerun at the pilot's size landed within 2.0 accuracy points of the proposal's table.
- **Real-data screen (Nhan).** 33 OpenML datasets screened with classic models only over 10 seeds (4,200 fits; 27 at n=100 and 500, 6 at n=100 only). TabPFN has not seen any of them.
- **Model access (Kiet).** TabPFN v2 and 3.5 both ran on a Colab A100 without errors, taking about 3.2 s and 4.8 s per full 200×200 grid job. This single-problem test gives a rough estimate of 4 GPU hours for the planned runs.
- **Flagged for the metric freeze.** Near-flat surfaces leave shape metrics undefined in some classic runs (248 for axis alignment, 15 for jump share), and SVM probabilities are unreliable at very small n.

## Response to Feedback

| Feedback | Decision | What changed |
|---|---|---|
| 1. Step score | Replaced | S is renamed axis alignment; jump share now measures sharpness. Jump share depends on its threshold (random forest: 0.94 at 0.01, 0.36 at 0.1), so we will fix the value before reading sweep results and report sensitivity. |
| 2. Statistical strength | Accepted | 10 seeds, with paired differences, 95% CIs, effect sizes and pre-set margins. Inconclusive results are reported as such. |
| 3. Two datasets | Both options | Framed as a case study, with roles assigned using classic models only. Proposed: covertype, run_or_walk, kdd_ipums_la_97_small (tree-favoring) and california, eeg_eye_state, kc1 (smooth-favoring). Smooth-favoring data proved scarce, so we relaxed that side's rule after seeing classic results, before TabPFN, and will disclose this. eeg_eye_state needs a re-screen with time-block splits. Fallback: a smaller case study. |
| 4. Classmate study | Removed | Replaced by 3–5 expert walkthroughs reported qualitatively, plus our worked examples. |
| 5. Visualization claim | Accepted | D3 only; Streamlit removed. Claim: a slider shows one boundary evolving as context grows, and compare mode shows $\lvert pA - pB \rvert$, with a worked example of each. A static page cannot run TabPFN, so the slider uses precomputed nested contexts. |
| 6. Model access | Checked | Go for H5, H6 and RQ3, on the result above. TabPFN runs remain on Kiet's Colab. |

## Scope and Plan

**Completed:** the foundation, metric test, classic sweep, dataset screen and model access check above. **Remaining:** TabPFN sweeps, real-data tests, statistical analysis, the D3 interface, walkthroughs and the paper.

| Dates | Kiet | Nhan |
|---|---|---|
| Nov 4–10 | Freeze metrics, hypotheses and margins; run TabPFN synthetic sweep, results unread until the freeze | Finalize datasets and re-screen eeg_eye_state; D3 canvas and controls |
| Nov 11–17 | RQ1 and RQ3 analysis; paper skeleton | Small multiples and metrics panel |
| Nov 18–24 | TabPFN real-data runs; RQ2 analysis | Real-data pipeline and baselines; slider and compare mode |
| Nov 25–Dec 1 | Full paper draft; walkthroughs | Story mode; walkthroughs |
| Dec 2–8 | Freeze results Dec 5; revise paper | Clean-clone reproducibility test; polish interface; revise paper |
| Dec 9–12 | Final fixes; submit by Dec 11 | Final fixes; Dec 12 is buffer |

**If time runs short, we cut in order:** ensemble-size ablation, noise-column test, H6, the additional datasets, story mode.

## GitHub Repository

https://github.com/KNguyen37/tabpfn-inductive-bias (public; the README lists how to reproduce current results)

## AI Usage Disclosure

The project's direction and research ideas originated with us. We used Claude (Claude Code) to help with coding, experiment orchestration, writing and organization. It also suggested some technical options, such as the jump share metric, which we reviewed before adopting. We are responsible for all content and results.